🎓 Student Performance Analysis | Power BI

📌 Project Overview

The Student Performance Dashboard is an interactive Business Intelligence solution developed in Power BI to analyze student academic performance across Mathematics, Reading, and Writing.

The dashboard provides educational stakeholders with data-driven insights into academic outcomes and highlights the influence of gender, parental education level, test preparation participation, and ethnic group on student achievement.

This project demonstrates end-to-end Business Intelligence capabilities, including data cleaning, transformation, data modeling, DAX calculations, dashboard development, and insight generation.

🎯 Business Problem

Educational institutions require accurate and accessible performance reporting to understand student achievement patterns and identify factors influencing academic success.

Without analytical reporting, it becomes difficult to:

- Monitor overall student performance.
- Identify underperforming student groups.
- Evaluate the effectiveness of test preparation programs.
- Assess the influence of parental education on academic achievement.
- Make informed educational decisions.

This dashboard was developed to transform raw student performance data into actionable insights for educators, administrators, and decision-makers.

🎯 Project Objectives

The project aims to:

- Analyze overall student performance.
- Compare Mathematics, Reading, and Writing achievement.

- Evaluate performance differences across genders.
- Assess the impact of parental education levels.
- Measure the effectiveness of test preparation programs.
- Examine academic outcomes across ethnic groups.
- Generate actionable recommendations for improving student success.

🛠 Tools & Technologies Used

Power BI

Used for:

- Dashboard Development
- Interactive Data Visualization
- KPI Monitoring
- Report Design

Power Query

Used for:

- Data Cleaning
- Data Transformation
- Data Preparation
- Data Quality Validation

Data Cleaning Activities

- Checked for missing values
- Removed inconsistencies
- Standardized categorical values
- Verified data types
- Prepared data for analysis

DAX (Data Analysis Expressions)

Used to create analytical measures and KPIs.

Key DAX Measures

- Total Students

- Overall Average Score
- Average Math Score
- Average Reading Score
- Average Writing Score

📊 Dashboard KPIs

KPI	Value
Total Students	100
Overall Average Score	61.18%
Average Math Score	62.51
Average Reading Score	60.49
Average Writing Score	60.53

📈 Dashboard Analysis

1. Overall Student Performance

The dashboard reveals an overall average score of **61.18%**, indicating moderate academic performance across the student population.

Insight

- Student performance is above average but still leaves significant room for improvement.
- Academic support initiatives can further improve overall outcomes.

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2. Subject Performance Analysis

Subject	Average Score
Mathematics	62.51
Reading	60.49
Writing	60.53

Insight

- Mathematics achieved the highest average score.
- Writing performed slightly better than Reading.
- Reading recorded the lowest average performance among all subjects.

Business Interpretation

Students appear to demonstrate stronger quantitative skills than literacy-related competencies.

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3. Performance by Gender

The dashboard evaluates academic outcomes across gender groups.

Insight

- Performance differences between genders are relatively small.
- Both gender groups demonstrate similar academic achievement levels.
- Gender does not appear to be a major determinant of academic success within the dataset.

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4. Performance by Parental Education Level

Student performance was analyzed against parental educational attainment.

Insight

- Students whose parents possess higher educational qualifications generally perform better academically.
- Academic performance improves progressively with parental education level.

Business Interpretation

Parental educational attainment may positively influence student learning support and academic engagement.

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5. Test Preparation Course Impact

The dashboard compares students who completed a test preparation course against those who did not.

Insight

- Students who completed test preparation courses consistently achieved higher academic scores.
- Test preparation demonstrates a measurable positive impact on student performance.

Business Interpretation

Test preparation programs contribute significantly to improved academic outcomes.

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6. Performance by Ethnic Group

The dashboard examines academic outcomes across ethnic groups A through E.

Insight

- Performance levels vary across ethnic groups.
- Certain groups consistently outperform others.
- Achievement gaps exist across demographic segments.

Business Interpretation

Targeted educational support may be required to address performance disparities.

🔍 Key Insights

Academic Performance

- Overall student performance stands at 61.18%.
- Mathematics is the strongest-performing subject.
- Reading is the weakest-performing subject.

Student Demographics

- Gender-based performance differences are minimal.
- Performance variation is more strongly associated with educational and demographic factors.

Educational Factors

- Test preparation participation improves academic outcomes.
- Students from households with higher parental education levels generally perform better.

Equity and Inclusion

- Performance disparities exist among ethnic groups.
- Additional support may be required for lower-performing groups.

Recommendations

1. Strengthen Reading Programs

Since Reading recorded the lowest average score, schools should:

- Implement reading intervention programs.
- Increase literacy development activities.
- Introduce targeted comprehension support.

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2. Expand Test Preparation Initiatives

Given the positive relationship between preparation and performance:

- Encourage greater participation in preparation courses.
- Provide access to practice assessments and study resources.
- Integrate preparation support into academic programs.

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3. Support Underperforming Groups

Develop targeted intervention strategies for demographic groups exhibiting lower academic achievement.

Examples include:

- Tutoring programs
- Mentorship initiatives
- Academic coaching

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4. Increase Parent Engagement

Schools should strengthen parent-student engagement through:

- Academic workshops
- Progress monitoring sessions
- Learning support resources

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5. Implement Continuous Performance Monitoring

Leverage dashboards and analytics to:

- Track student performance trends.
- Identify performance gaps early.
- Support evidence-based decision making.

🗁 Project Workflow

Phase 1: Data Collection

- Imported student performance dataset into Power BI.

Phase 2: Data Cleaning & Preparation

Using Power Query:

- Validated data quality.
- Standardized fields.
- Prepared analytical dataset.

Phase 3: Data Modeling

- Structured data model.
- Optimized relationships.
- Prepared measures for analysis.

Phase 4: DAX Calculations

Created measures for:

- Student Count
- Subject Averages
- Overall Performance Percentage

Phase 5: Dashboard Development

Developed interactive visualizations including:

- KPI Cards
- Column Charts
- Bar Charts
- Donut Charts
- Interactive Filters

Phase 6: Insight Generation

Analyzed:

- Subject Performance
- Gender Performance
- Parental Education Impact
- Test Preparation Effectiveness
- Ethnic Group Performance

📊 Business Impact

This dashboard enables educational stakeholders to:

- Monitor academic performance efficiently.
- Identify student achievement gaps.
- Evaluate educational support programs.
- Improve resource allocation.
- Make data-driven academic decisions.

🚀 Skills Demonstrated

- Data Cleaning
- Data Transformation
- Power Query
- Data Modeling
- DAX Calculations
- Data Visualization
- Dashboard Design
- Business Intelligence
- Data Storytelling
- Insight Generation
- Stakeholder Reporting

👤 Author

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